

KWK Applied Data Science Fundamentals Challenge

TOPIC:

CYBERSECURITY

SUMAYYA MARIYAM

```
obj.1): "translator": null  
obj.1): "protector": null  
obj.1): "verifier": null  
obj.1): "followers_count": 0  
obj.1): "friends_count": 0  
obj.1): "listed_count": 0  
obj.1): "favourites_count": 0  
obj.1): "statuses_count": 0  
obj.1): "created_at": "2015-07-17T22:31:00Z"  
obj.1): "utc_offset": null  
obj.1): "time_zone": null  
obj.1): "geo_enabled": true  
obj.1): "lang": "en"
```

01.

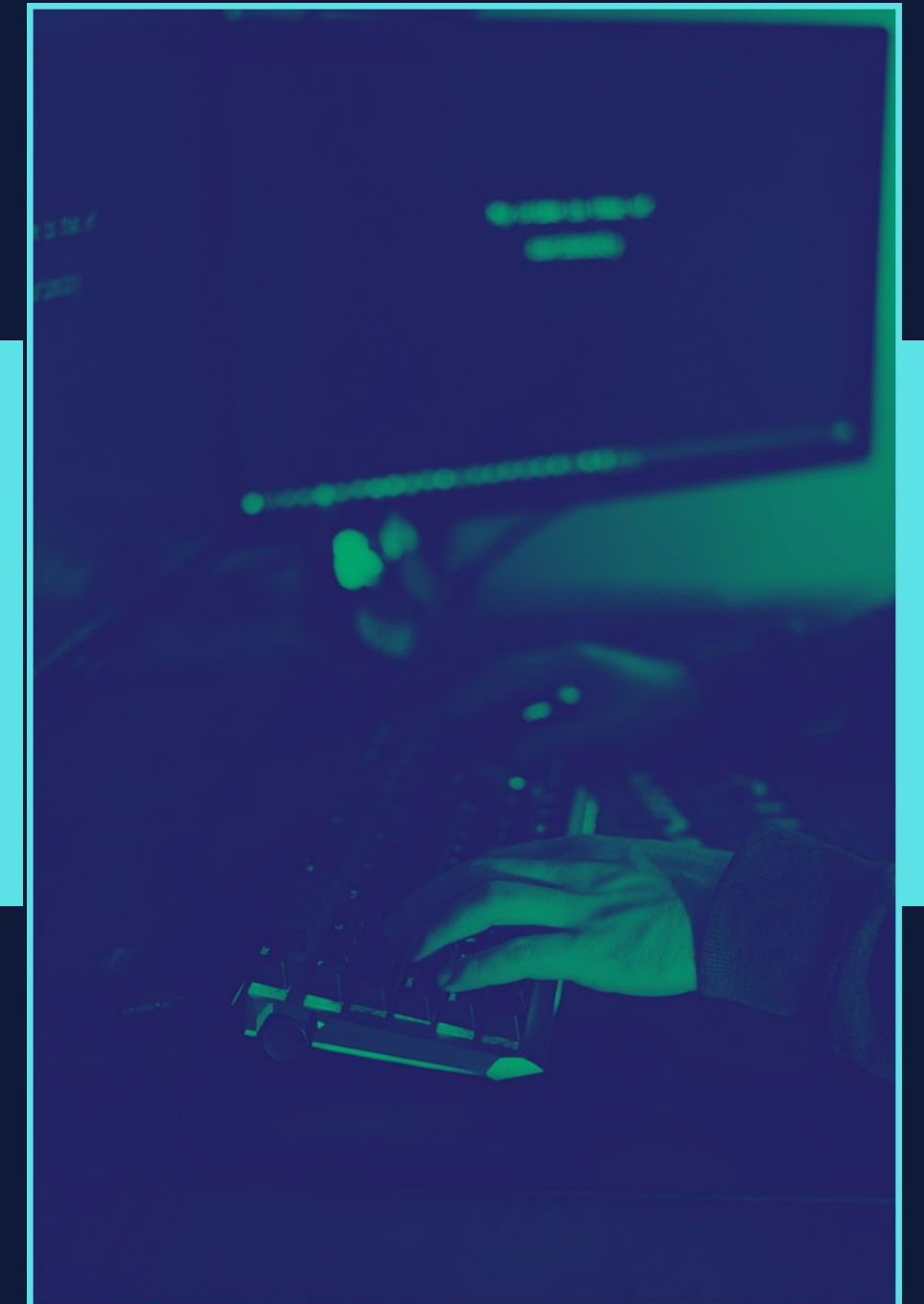
Problem Statement

Over the years cyberattacks have continued to grow, they have become more frequent and extremely sophisticated costing businesses and organizations billions of dollars while affected millions of victims who have their sensitive data exposed. Because of the nature of cyber attacks, thousands of causes are documented every year but it's not always clear which vulnerabilities are most commonly exploited and which attack methods adversaries favor.

Finding my way through this challenge I plan to seek answers to two key questions:

- **What are the most common system vulnerabilities that are targeted by cybercriminals?**
- **What attack methods are most favored by adversaries to exploit these vulnerabilities?**

By analyzing cybersecurity incident and vulnerability datasets, the goal of this project is to find trends in how attackers operate. By understanding these patterns, we can be proactive about future cyber attacks by prioritizing defense strategies, improve detection of anomalies, and allocate our resources effectively to prevent million to billion dollar losses. This is especially important considering the risk of cyberattacks is only projected to over the upcoming years



02. Where is my data from?

According to my short research, Kaggle had the type of datasets I was looking for and I am allowed to use their data. And a perk each dataset has a 'usability' score that determines variables like Completeness, Credibility, and Compatibility so I can have insights on my datasets.



04.

How was the data cleaned?

Using the resources provided, I decided to use Python libraries to clean my datasets: Pandas and Numpy and matplotlib

I dropped all unnecessary columns and removed any rows with missing data

```
rows = rows[1:] # removes headers, unnecessary row
dic = {} # keep track of unique attack typea and their counts
# print(rows[0])

• for row in rows:
    # print(row[3].strip())
    if row[3].strip() not in dic: # row[3] holds attack types
        dic[row[3].strip()] = 0 # cleans data, removes any extra whitespace
    dic[row[3].strip()] += 1 # keep track of # of occurence

# for key, value in dic.items():
#     print(key, ":", value)
dic['other'] = 0 # cleans irrelevant data
keys = []

• for key, value in dic.items():
    if value <= 30 : # if an attack occurs 30 times or less, its too rare to be most common
        dic['other'] += 1
        keys.append(key)

for key in keys: # remove rare attack types
    del dic[key]
del dic['other']
```

04.

How were visualizations created?

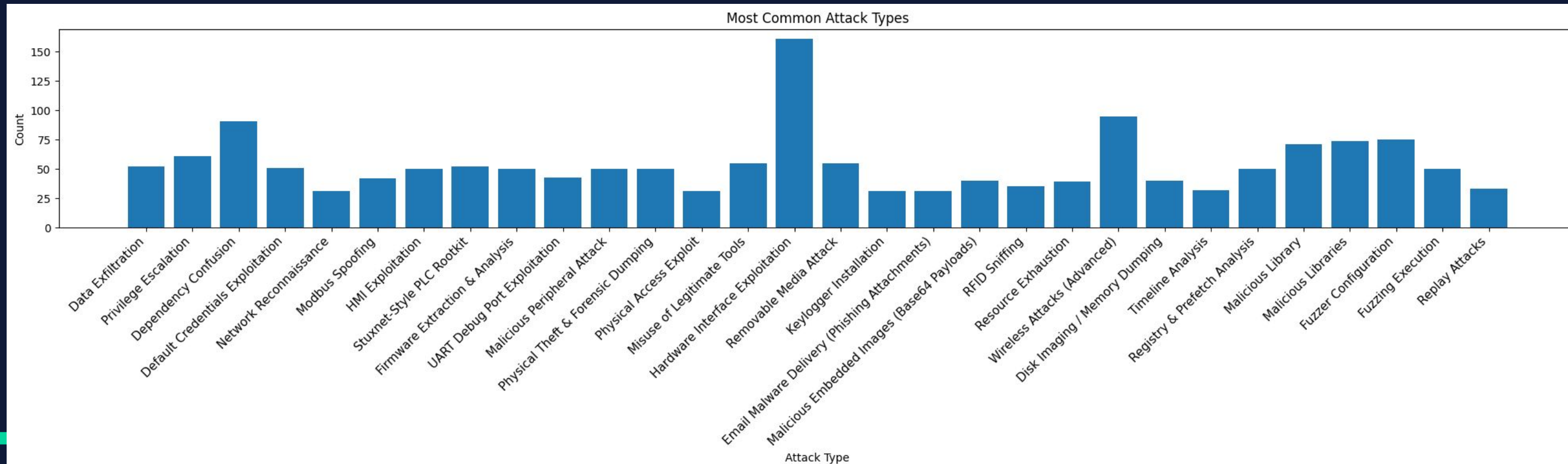
Using the resources provided, I decided to use Python libraries to clean my datasets: Pandas and Numpy and matplotlib

I added the graph bars based off dictionary values, and labeled everything to be analyzable

```
# plot using matplotlib
print(dic)
plt.figure(figsize=(20, 6)) # width=20, height=6
plt.bar(dic.keys(),dic.values())
plt.xlabel("Attack Type")
plt.ylabel("Count")
plt.title("Most Common Attack Types")
plt.xticks(rotation=45, ha="right", fontsize=11)
plt.tight_layout()
plt.show()
```

FINDINGS

MOST COMMON ATTACK TYPES

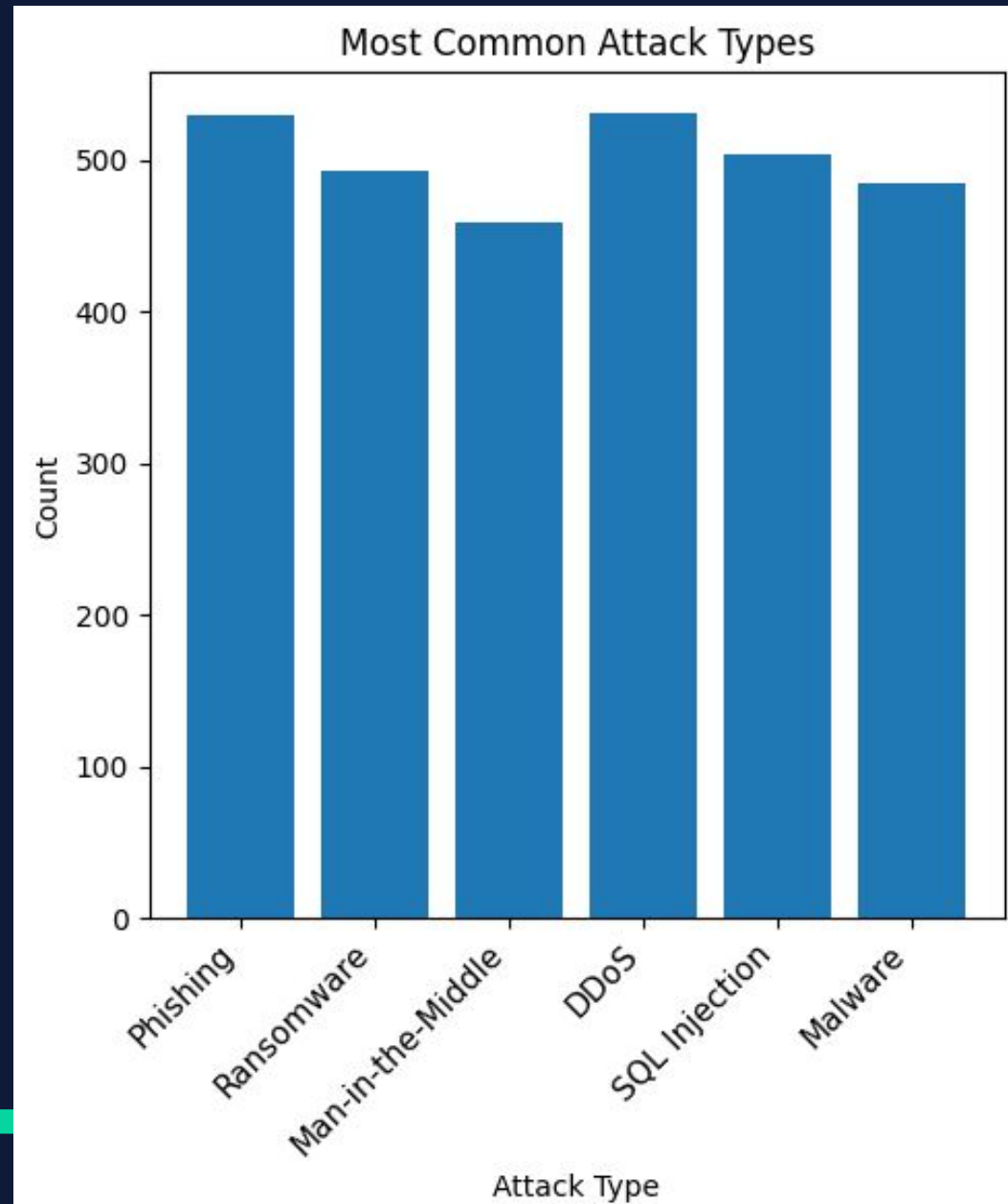


Dataset: Attack_Dataset.csv

Conclusion: Most common attack types seem to be Dependency Confusion, Hardware Interface Exploitation, and Wireless Attacks

FINDINGS

MOST COMMON ATTACK TYPES



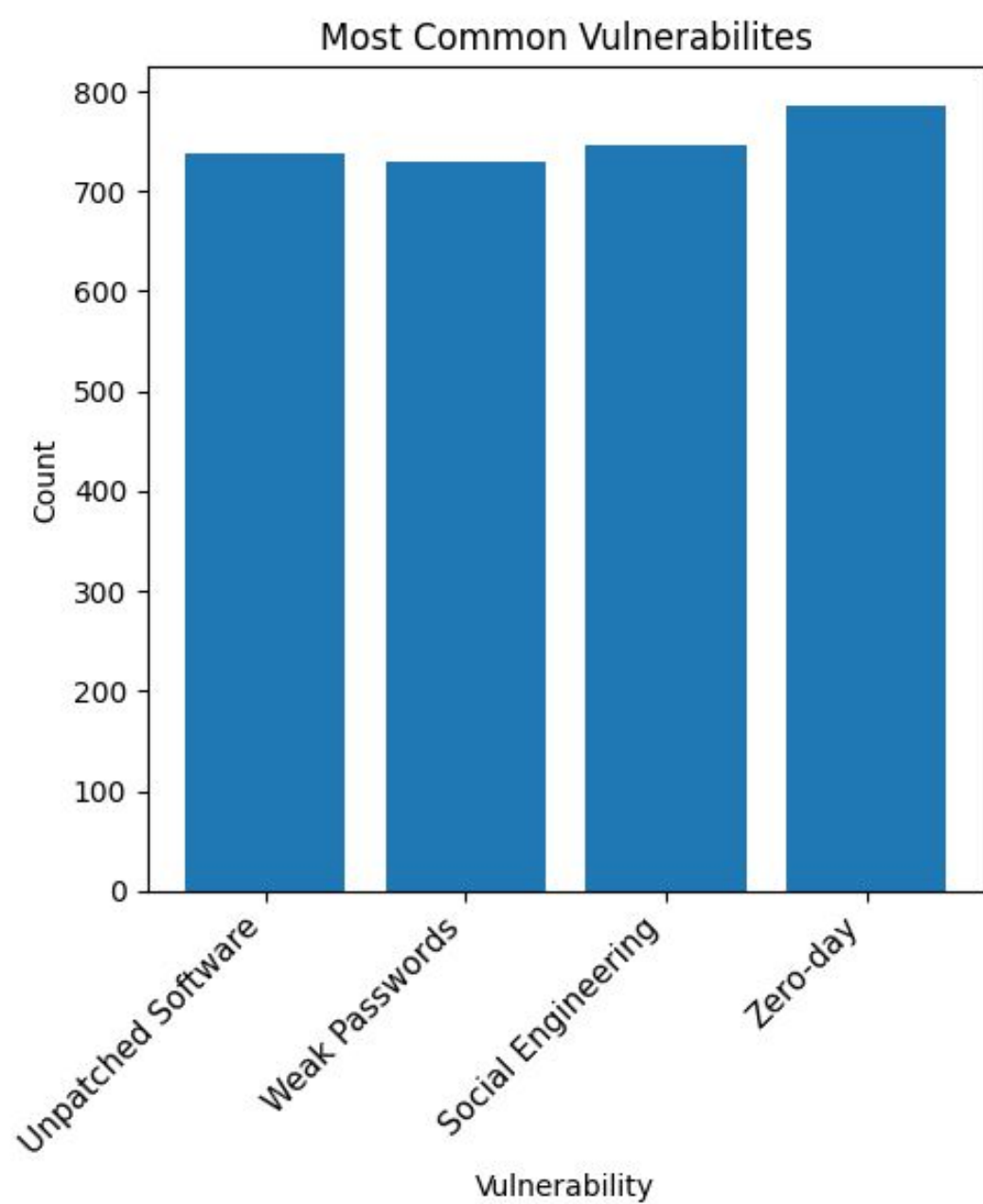
Dataset:

Global_Cybersecurity_Threats_2015-2024.csv

Conclusion: Most common attack types on a global scale seem to be Phishing and DDoS

FINDINGS

MOST COMMON VULNERABILITY TYPES



Dataset:

Global_Cybersecurity_Threats_2015-2024.csv

Conclusion: Most common vulnerability types on a global scale seem to be unpatched software and zero-day.

04.

How AI helped better visualize data.

AI suggested I use the seaborn library to create heatmaps leveling up my visualizations.

After cleaned my data I used seaborn's heatmap function to better understand correlations between attacks and vulnerabilities

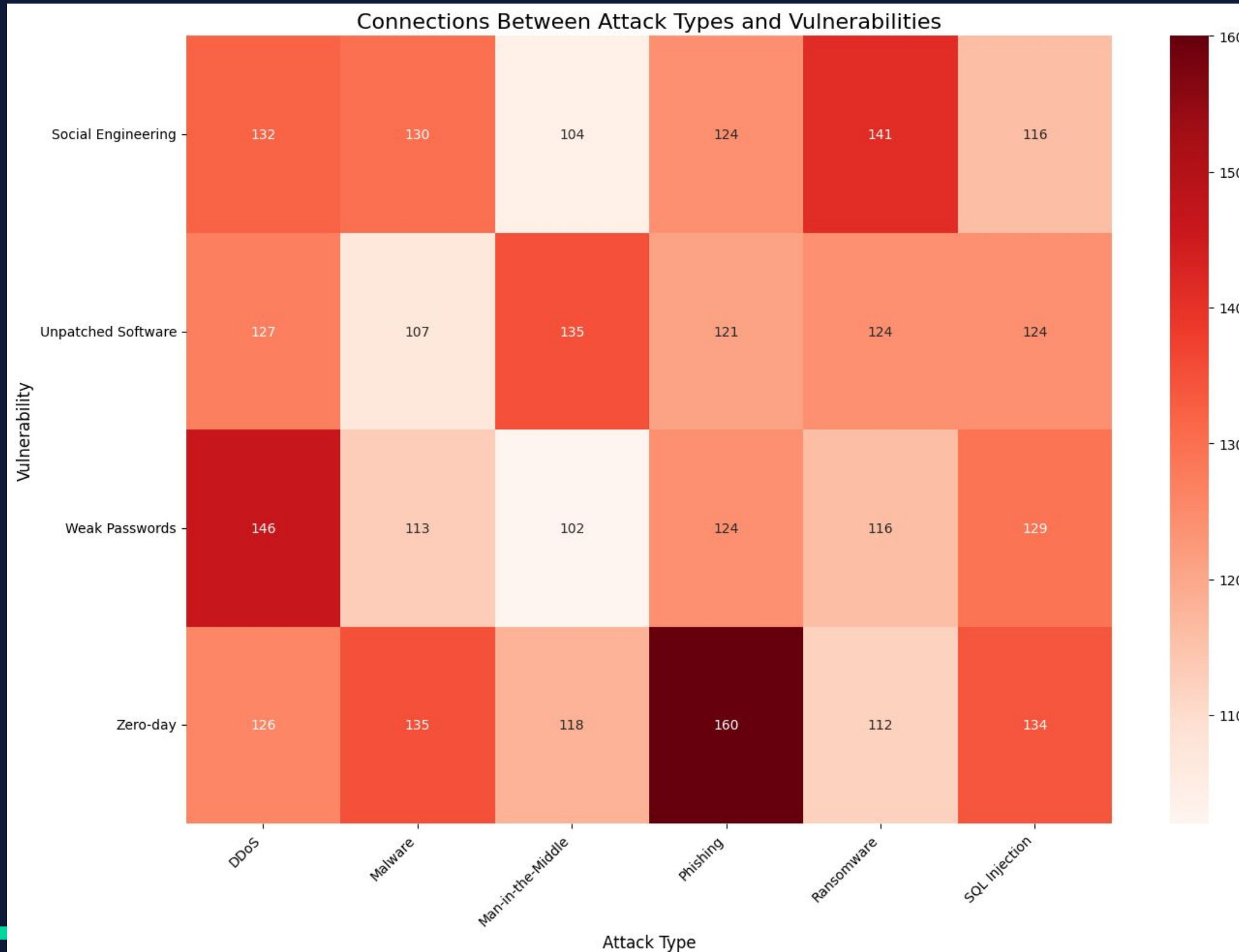
```
# clean data, remove whitespace
df["Attack Type"] = df["Attack Type"].str.strip()
df["Security Vulnerability Type"] = df["Security Vulnerability Type"].str.strip()

# pivot table for all attack types vs vulnerabilities
pivot = pd.pivot_table(df,
                        index="Security Vulnerability Type",
                        columns="Attack Type",
                        values="Country", # any column to count total value num
                        aggfunc="count",
                        fill_value=0)

# plot heatmap
plt.figure(figsize=(14, 10))
sns.heatmap(pivot, cmap="Reds", annot=True, fmt="d", cbar=True)
# cmap="Reds": draker red intensity = higher frequency
# annot=True: shows numbers inside cells.
# fmt="d": numbers formatted as integers.
```

FINDINGS

CAUSES IN RELATIONS TO ATTACKS



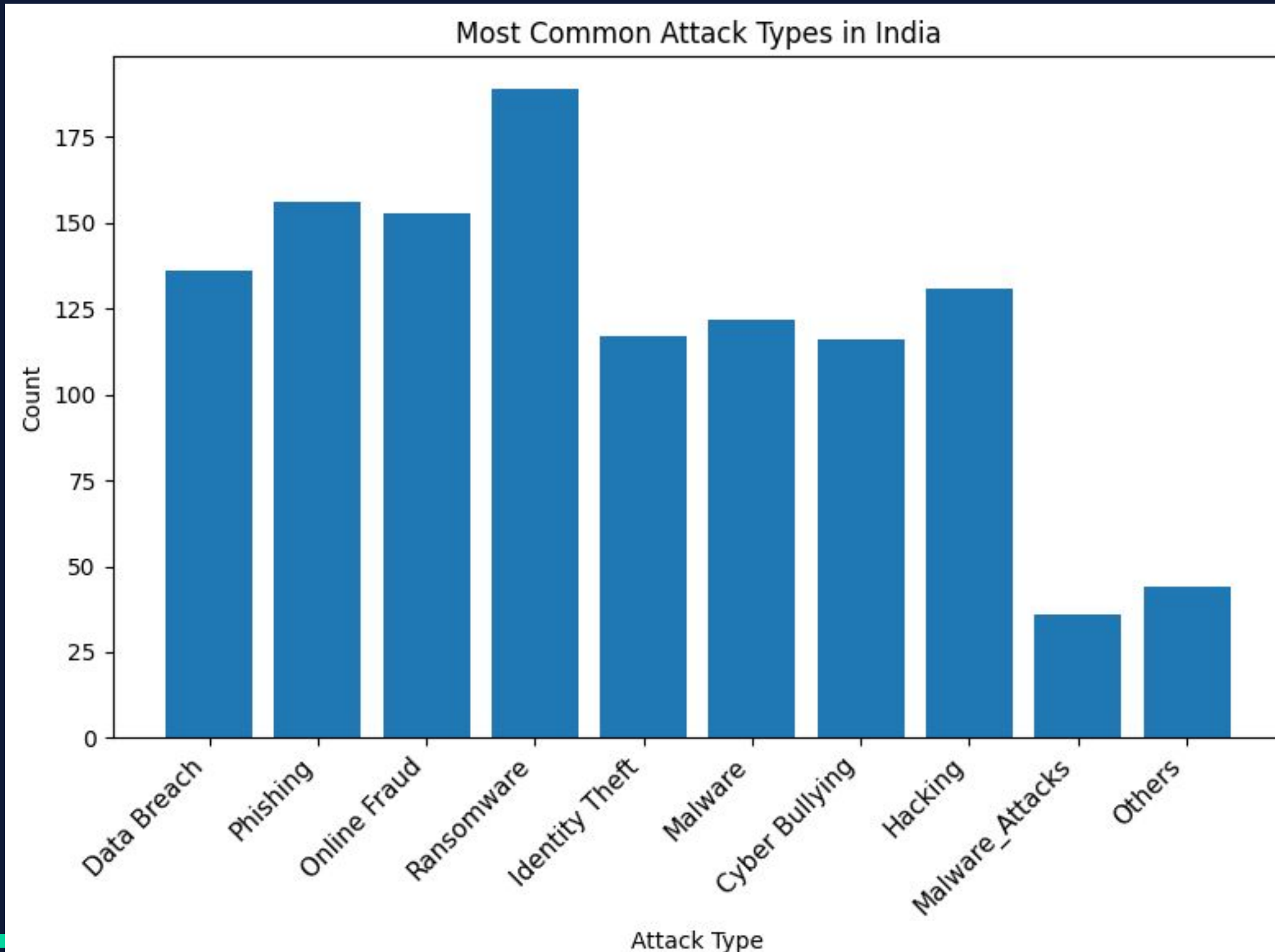
Dataset:

Global_Cybersecurity_Threats_2015-2024.csv

Conclusion: We are able to see some obvious vulnerabilities that lead to certain attack types: Zero-day vulnerability leads to Phishing attack, Weak Passwords lead to DDoS attacks.

FINDINGS

MOST COMMON ATTACK TYPES: INDIA



Dataset:

cybersecurity_cases_india_combined.csv

Conclusion: In India specifically we see Ransomware and Phishing as most common attack types

CONCLUSIONS

Most Frequent Vulnerabilities: Cybercriminals' most frequent vulnerabilities to exploit are unpatched software, zero-day vulnerabilities, and weak passwords. They are widespread and have the ability to allow attackers to gain access or crash systems with ease.

Most Common Attack Methods: Globally, Phishing and DDoS attacks are the most common ones, followed by other common methods such as Ransomware, Dependency Confusion, Hardware Exploitation, and Wireless Attacks. These attacks hit both technical vulnerabilities and human nature.

Vulnerability-Attack Similarity: Certain of the vulnerabilities strongly correlate with certain attacks:

Zero-day vulnerabilities → Phishing attacks

Weak passwords → DDoS attacks

CONCLUSIONS

We can zoom in and analyze regional trends: phishing and ransomware are India's most common attacks, reflecting local cybercrime trends. This regional pattern suggests that targeted awareness campaigns, backup strategies, and other security measures are particularly critical in reducing local exposure to financially motivated attacks. Focusing on regional trends is also crucial as it's important to understand each region has different widely available resources, in countries where adversaries may not have access to more sophisticated attack methods, global trends may not be helpful in risk analysis.

Insights:

Organizations can be proactive to fix high-risk vulnerabilities, improve detection, and prioritize defenses by knowing about these trends.

Security teams can prioritize their resources better by correlating vulnerabilities with the attacks they enable, and avoid mass data breaches and financial losses.

GITHUB REPOSITORY:

<https://github.com/smariyamm/KWK-2025-DataScience-Fundamentals-Challenge-SM>